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1. Project Description

For this assignment I have created a virtual campsite in OpenGL where the user can use W and S to move forward and backwards based on the direction they are looking and A and D to change where they are looking. The Scene includes 150 instances of a tree I created manually by combining a brown cylinder with hard coded vertices, normals and indices with 4 green cones varying in size and shade stacked and overlapping each other. Four obj objects were loaded from the internet, including a campfire, 5 chairs surrounding the campfire, a picnic table off to the side and finally a tent on the other side. The user is walking on a simple green square plane and lastly the sky was made a light blue.

2. Environment data structure and rendering

The scene is made up of a bunch of VAOs. Each imported model uses a separate VAO and associated vertex and index buffers. The tinyobjectloader from our in class example, reads the OBJ files and the program then finds the object's lowest point and center point to then apply an offset based on the coordinates at which it will be placed when rendering.. The trees however, as mentioned above were built from reusable hard coded trunk geometry made of 8 vertices at the bottom of the cone and 8 at the top connected with two triangles for each side of the cylinder leaving the top and bottom unfilled since they would be covered by the ground and leaves. The cone geometry was made more simply, using a loop to build a single reusable cone mesh from eight triangular sides. Each tree uses one trunk and 4 cones with varying base heights and radii to create the look of a christmas tree. I chose separate VAO’s and buffers for the imported objects so that I could import all the objects independently but still use the same rendering process. For the repeated objects like the trees and chairs, I used the same geometry for each individual instance but stored their position in an array to prevent storing the same geometrical information more than once. This also made adjusting how many trees I wanted in the scene a simpler task.
For each frame, the program draws the ground plane then applies model, view, and projection transformations based on the current camera position to render the scene from the user’s new perspective.

3. Constraints

The objects are loaded using uniform colors due to time constraints. All the obj object’s mtl files are loaded in the repo however their textures are not loaded. This would be a good addition to the current project to make for a better looking scene. Another constraint that could be improved upon is the scene boundaries. Currently the scene just produces the plane, its objects and a blue background. Adding more details to the edges of the scene to make it look less like an endless abyss could also prove to be beneficial.